3. Using a calculator, find all solutions of

(*a*) $903x + 731y = 2107$;

(*b*) $903x + 731y = 1106$;

$(903, 731) = 43$ $\qquad (x_0, y_0) + k(\underbrace{17}_{\frac{731}{43}}, \underbrace{-21}_{\frac{903}{43}})$

(i)

903	731	2107
1		1
	1	1

$\Rightarrow$

172	731	2107
1		1
−1	1	1

$\Rightarrow$

172	43	2107
1	−4	1
−1	5	1

$\Rightarrow$

0	43	2107
17	−4	1
−21	5	1

$(x, y) = 49(-4, 5) + k(17, -21)$

(ii) No solution

because $\frac{1106}{43} = 25.9209\ldots$

$\notin \mathbb{Z}$

2021

−10585

구하는 문

5. Prove that $101x + 37y = 3819$ has a positive solution in integers.

- $1 = (101, 37) \mid 3819$ therefore solution exists

- $0 < x_1 + kb$ $\qquad -\frac{x_1}{b} < k < \frac{y_1}{a}$

  $0 < y_1 - ka$

  $ax_1 + by_1 = c$ $\qquad$ 주간의 길이 : $\frac{x_1}{b} + \frac{y_1}{a} = \frac{c}{ab}$.

- $\left| p - \frac{c}{ab} \right| < 1$ where p = 경우의수의 개수

- $\frac{3819}{101 \cdot 37} = 1.02194\ldots > 1$ therefore $p > 0$

Theorem 5.1 *Let a, b and c be integers with not both a and b equal to 0, and let $g = \text{g.c.d.}(a, b)$. If $g \nmid c$ then the equation (5.1) has no solution in integers. If $g \mid c$ then this equation has infinitely many solutions. If the pair (x_1, y_1) is one integral solution, then all others are of the form $x = x_1 + kb/g$, $y = y_1 - ka/g$ where k is an integer.*

$-\frac{g}{b}x_1 < k < \frac{g}{a}y_1$

$p = -\left[-\frac{g}{a}y_1\right] - \left[-\frac{g}{b}x_1\right] - 1$

- least value $\left[-\frac{g}{b}x_1\right] + 1$

- greatest value $-\left[-\frac{g}{a}y_1\right] - 1$

$$\begin{bmatrix} 11 & 13 & 17 \\ 50 & 44 & 20 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 19 \\ 10 \end{bmatrix}$$

$$\begin{matrix} 11 & 13 & 17 \\ 50 & 44 & 20 \\ 1 & & \\ & 1 & \\ & & 1 \end{matrix} \Rightarrow \begin{matrix} 11 & 2 & 6 \\ 50 & -6 & -30 \\ 1 & -1 & -1 \\ & 1 & \\ & & 1 \end{matrix} \Rightarrow \begin{matrix} 11 & 2 & 0 \\ 50 & -6 & -12 \\ 1 & -1 & 2 \\ & 1 & -3 \\ & & 1 \end{matrix}$$

$$\Rightarrow \begin{matrix} 1 & 2 & 0 \\ 80 & -6 & -12 \\ 6 & -1 & 2 \\ -5 & 1 & -3 \\ & & 1 \end{matrix} \Rightarrow \begin{matrix} 1 & 0 & 0 \\ 80 & -166 & -12 \\ 6 & -13 & 2 \\ -5 & 11 & -3 \\ & & 1 \end{matrix}$$

$$\Rightarrow \begin{matrix} 1 & 0 & 0 \\ 80 & -10 & -12 \\ 6 & -39 & 2 \\ -5 & 50 & -3 \\ & -13 & 1 \end{matrix} \Rightarrow \begin{matrix} 1 & 0 & 0 \\ 80 & -10 & -2 \\ 6 & -39 & 41 \\ -5 & 50 & -53 \\ & -13 & 14 \end{matrix}$$

$x = 19$

$y = -5$.

$$\Rightarrow \begin{matrix} 1 & 0 & 0 \\ 0 & 0 & -2 \\ 1646 & -244 & 41 \\ -2125 & 315 & -53 \\ 560 & -83 & 14 \end{matrix} \left| \begin{matrix} 19 \\ 10 \end{matrix} \right.$$

$$19 \cdot (1646, -2125, 560) - 5(41, -53, 14) + 1 \cdot (-244, 315, -83)$$

- integer lattice $\Lambda = \{ c_1 \underbrace{v_1}_{row} + c_2 \underbrace{v_2}_{row} + \cdots + c_m \underbrace{v_m}_{row} \}$

- $\underbrace{U}_{\text{가역 정수 성분}} M = \underbrace{H}_{\text{어느에르 표준형.}}$ $\longrightarrow$ U는 역행렬의 조상들 이어주기 함으로 H는 Λ의 기저가 됨.

  $\det U = 1$

- $h_i = (0, 0 \cdots 0, d_i, \times \times \times)$

  $d_i > 0$

- let Λ_i : 높이 i번째 i-1번째까지가 0인 벡터들의 집합.

$$= \{ \underbrace{k_i h_i}_{\substack{i\text{번째} \\ \text{성분이 } d_i}} + \underbrace{k_{i+1} h_{i+1}}_{\substack{i\text{번째 성분이} \\ \text{모두 } 0}} + \underbrace{\cdots}_{} \}$$

$\Rightarrow$ $d_i \mid \Lambda_i$의 모든 벡터의 i번째 성분

$\Rightarrow$ $d_i = \Lambda_i$이 가질 수 있는 i^{th} 성분 중 가장 작은 양의 정수.

$\rightarrow$ d_i는 양변 단위 증가라 구분하기 정하기 ,